Course Name: Final Exam Sheet

Open-book reference

Student: Your Name (YOUR-ID)

Exam playbook (read this first)

What the teacher announced, and any hints (board photos, slides shown in class). Then a map from question type to page.

Question type	What can change	Go to
Type A	numbers, start point	Part 1, page 3

Habits that save marks.

- Write the general formula first, then plug in the numbers.
- Put iterations in a table: one row per iteration.
- After each step, check: did the objective go down? Is the point still allowed?

> **Hand-calculation helpers.** Put the small formulas you keep looking up here.

Part 0: The math you need, from zero

Read once before the exam.

0.1 A basic idea

Explain it in plain words, then give a number example, then the symbols.

Part 1: Topic name

Lecture N

Question 1: Short name

1 Question title as printed Pool §1, p. 1

Copy the question exactly as printed, with the same equation numbers.

$$\min_{\mathbf{x}} \; f(\mathbf{x}) = \tfrac{1}{2}x_1^2 + 2x_2^2. \tag{1}$$

(a) First part.

(b) Second part.

The idea in plain words. One short paragraph.

Step-by-step guideline works for any numbers

STEP 1 **Name of step**

What to do, written with letters.

STEP 2 **Name of next step**

What to do next.

Worked example

STEP 1 **Name of step**

The same step with the real numbers, every line shown.

STEP 2 **Name of next step**

…

ITERATION 1 at $x^{(0)} = (2, 1)$:

- Gradient: …

Traps

The sign mistake people make here, and how to check for it.

Answer: Question 1 what to write on the exam paper

(a) Short working, then the final answer in a box: $\boxed{x^\star = (1, 0)}$

(b) …